

Chapter 2

The Extensive Form

There are two standard ways to write a game down mathematically: the **extensive form** and the **normal form**.

This chapter develops the extensive form, which makes the timing of players' decisions explicit by laying them out as a tree.

We will build up the idea using a single running example, then look at a few more, and finish with an exercise for you to work through yourself.

The Elements of a Game

Any mathematical representation of a game, extensive or normal form, has to specify five things: the players, what each can do, what each knows when he acts, how their choices lead to an outcome, and how each player ranks the possible outcomes.

The rules of chess illustrate the first four elements precisely: there are two players; they alternate moving pieces subject to rules about legal moves; each player observes the entire history of play; and the game ends when a king is captured, or in a draw.

Preferences are not part of chess's rules, but we would normally assume that a player prefers winning to a draw, and a draw to losing.

Nodes and Branches

An extensive-form game is drawn as a tree, built out of **nodes** and **branches**.

Definition. A **node** represents a point in the game, most often a point at which one of the players has to make a decision.

Definition. A **branch** is an arrow connecting one node to another, labeled with the action that takes the game from the first node to the second.

To see how this works, we will build up a tree for a real strategic episode: the 1998 rivalry between Disney's *A Bug's Life* and DreamWorks's *Antz*, two animated films about insects that opened within months of each other.

Setting Up the Bug Movie Game

Jeffrey Katzenberg ran Disney's animation studio until he left in 1994, reportedly after being passed over for a promotion. He went on to found DreamWorks with Steven Spielberg and David Geffen.

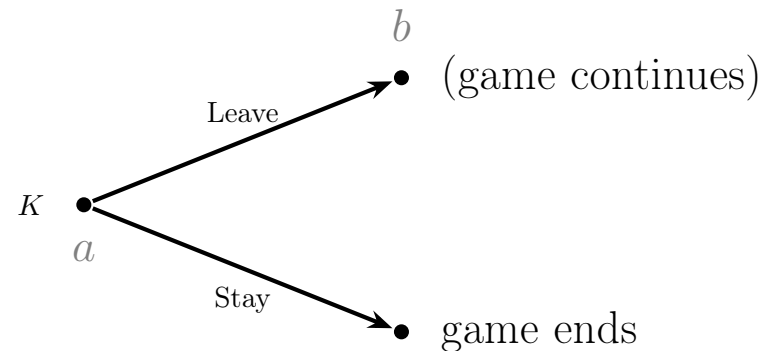
Shortly afterward, Pixar pitched Disney a movie about bugs. Around the same time, DreamWorks began developing its own bug movie, *Antz*, and rushed it into theaters ahead of Disney's *A Bug's Life*.

We will model this rivalry as a game between two players: Katzenberg (K) and Disney's Michael Eisner (E).



Katzenberg's First Move

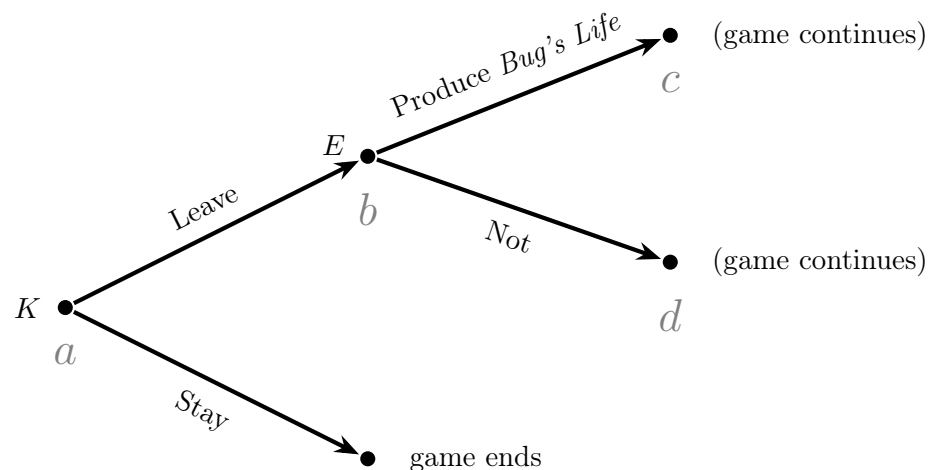
Every extensive-form game has exactly one **initial node**, where the game begins. Here, the game begins with Katzenberg's decision whether to leave Disney.



If Katzenberg stays at Disney, assume the game simply ends there. If he leaves, further decisions remain, so the tree keeps growing to the right.

Adding Eisner's Decision

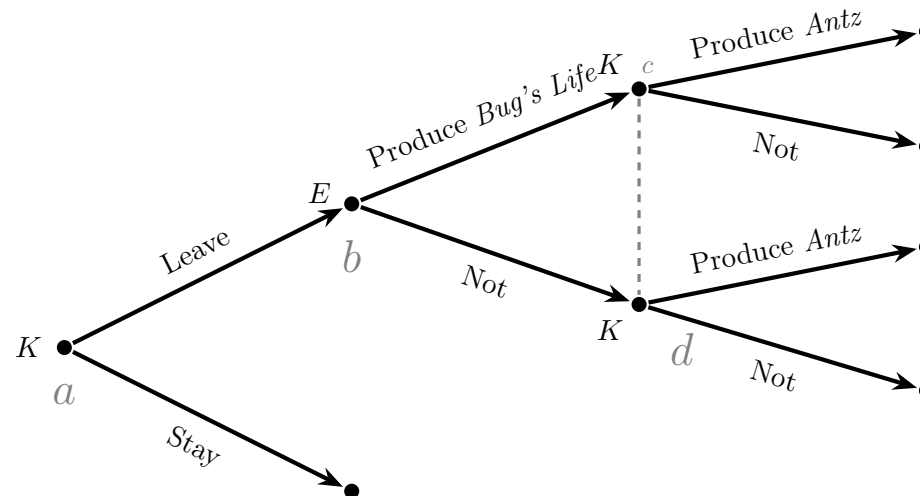
If Katzenberg leaves, Eisner must decide whether to produce *A Bug's Life*. Because Eisner only faces this choice after Katzenberg has left, his decision node sits at the node reached by the “Leave” branch.



At node b , Eisner knows he is at node b : he observed Katzenberg's move. This is worth stating explicitly, because it will not be true of every node we add next.

Adding Katzenberg's Second Decision

After Eisner's choice, Katzenberg must decide whether to produce *Antz*. But the two studios made this decision without observing each other: Katzenberg must choose before learning whether Eisner has committed to *A Bug's Life*.



The dashed line joining c and d marks that Katzenberg cannot tell them apart when he makes this second decision.

Information Sets

Definition. An **information set** collects the decision nodes that a player cannot distinguish between when it is her turn to move. Every decision node belongs to some information set; a set with only one node is called a singleton.

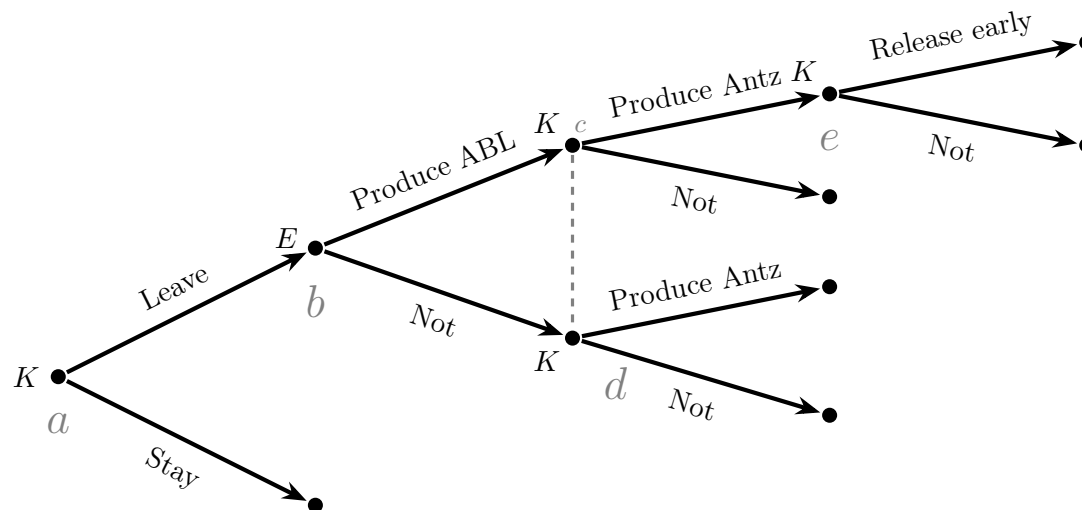
Node a is its own information set, since Katzenberg always knows he is making the first move. The same is true of node b for Eisner.

Nodes c and d , however, form a single non-singleton information set for Katzenberg. He knows he is at one of the two, but not which one, so he must make the same choice, produce *Antz* or not, at both.

We always assume every node inside one information set belongs to the same player, since it only makes sense to ask what a player knows when it is her turn to act.

The Full Tree

One decision remains: if both studios chose to produce, Katzenberg decides whether to rush *Antz* into theaters early, beating *A Bug's Life* to the screen. This is a new node, e , reached only along the “both produce” path.

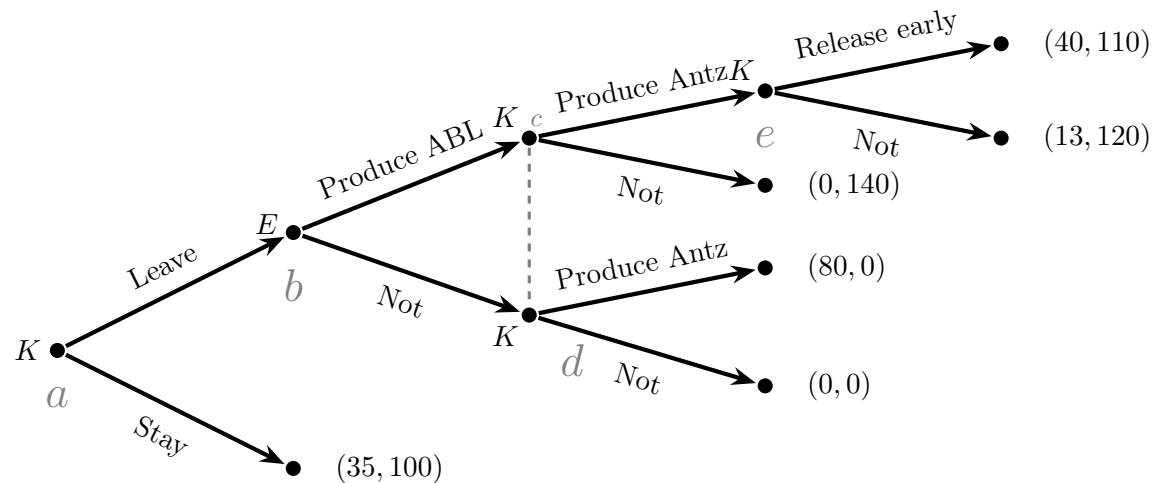


Decision nodes (a, b, c, d, e) are where a player moves. **Terminal nodes** are where the game ends: each one corresponds to exactly one path from the initial node.

Payoffs

Definition. A **payoff**, or utility, is a number representing how much a player values a particular outcome. Larger payoffs mean more preferred outcomes.

Suppose the profits (in millions of dollars) work out as follows, with Katzenberg's payoff listed first at every terminal node.



Quick Check: Reading the Tree

Looking back at the full tree on the previous slide, answer these without looking:

1. Which of Katzenberg's three information sets is *not* a singleton, and why?
2. If Eisner produces and Katzenberg matches him, who moves next, and what does she know at that point?
3. Suppose the game reaches a terminal node with payoff $(0, 140)$. Which path through the tree led there?

Answer: Reading the Tree

1. The $\{c, d\}$ set: Katzenberg cannot tell whether Eisner produced (node c) or not (node d) when he decides on *Antz*.
2. Katzenberg moves next, at node e ; he knows he is at e , since it is a singleton information set reached only along the “both produce” path.
3. Leave, then Produce *Bug's Life*, then Not produce *Antz*: the path $a \rightarrow b \rightarrow c \rightarrow (0, 140)$.

Compact Notation

Once a tree has many branches, it is convenient to abbreviate every action to a single letter, as in Figure 2.6 of Watson.

Katzenberg's choices are S (stay), L (leave), P (produce *Antz*), N (not), and, at node e , R (release early) or N' (not), using a prime to distinguish it from the earlier N .

Example. When Katzenberg is at node c or d , he only knows he must choose between P and N ; he does not know which node he is at.

Because c and d share one information set, we can use the same two labels, P and N , for both nodes' branches without any ambiguity about which action he is choosing.

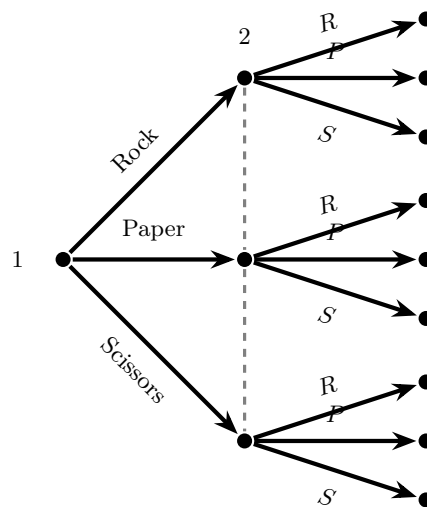
The Bug Movie Outcome

In the end, *Antz* earned a bit over \$90 million and *A Bug's Life* over \$160 million.

Whether Katzenberg and Eisner made the right decisions is exactly the kind of question the rest of this course will teach you to answer.

Example: Rock-Paper-Scissors

Information sets also give us a standard way to represent **simultaneous** play. In rock-paper-scissors, player 1 moves first in the tree; player 2 then chooses too, without having observed player 1's choice.



All three of player 2's nodes, reached via Rock, Paper, and Scissors respectively, share one information set: he cannot tell which he is at, the standard trick for representing simultaneous moves.

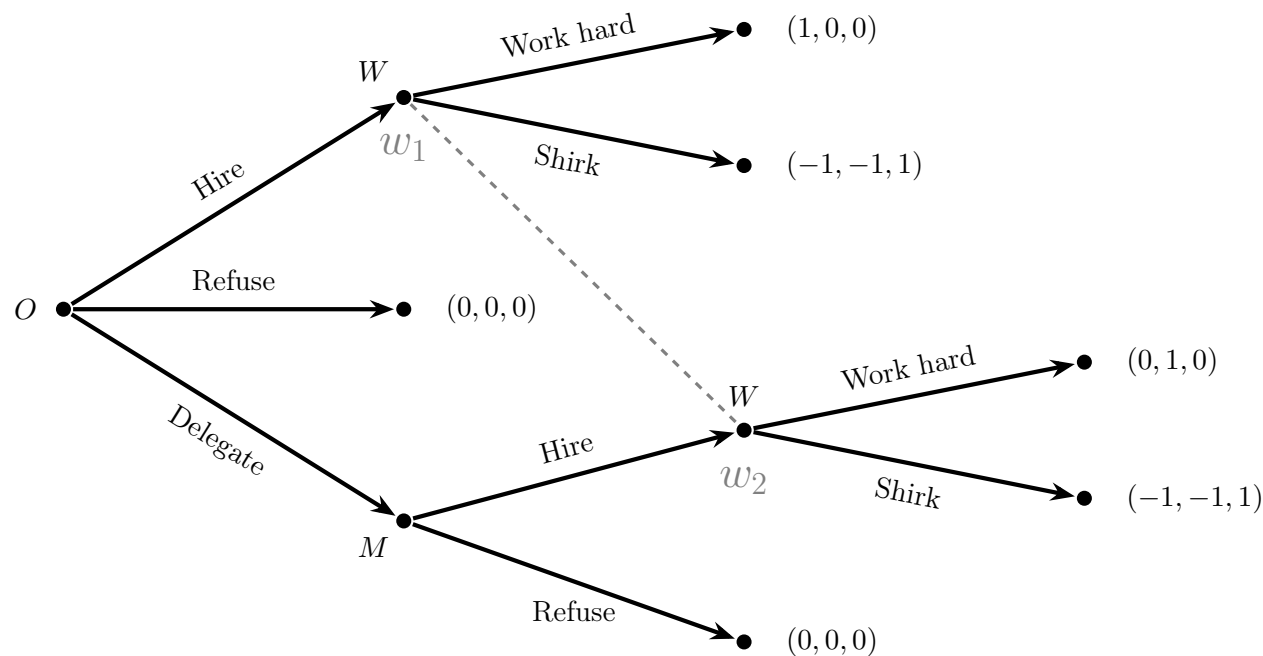
Discussion: Formulate a Game Tree

Consider the following strategic situation, adapted from Watson's Chapter 2 exercises.

- The owner of a Taipei restaurant decides whether to hire a new server, turn her away, or delegate the hiring decision to the floor manager.
- If hired, whether by the owner or by the manager, the server must decide whether to work hard or shirk, without knowing who hired her.
- If not hired, everyone receives a payoff of 0.
- If she shirks, she receives a payoff of 1, while the owner and the manager both receive -1 .
- If she works hard, whoever hired her receives a payoff of 1, and she and the other player each receive 0.

Draw and label the game tree, including the relevant information set.

Answer: The Owner–Manager–Worker Game



Payoffs are listed as (owner, manager, server). The dashed line is the key step: the server cannot tell whether she is at w_1 (hired by the owner) or w_2 (hired by the manager), so both nodes share one information set.

Chapter in a Nutshell

- An extensive-form game is a tree built from nodes and branches, with exactly one initial node.
- Decision nodes belong to information sets. A non-singleton information set means the player cannot tell which of those nodes she is at.
- Non-singleton information sets are the standard way to represent both forgotten past moves and simultaneous moves.
- Terminal nodes end the game and carry a payoff vector, one number for each player, with larger numbers meaning more preferred outcomes.

Key Concepts

- Node, branch, and initial node describe the basic building blocks of a game tree.
- Decision node and terminal node distinguish points where a player still has to act from points where the game has ended.
- Information set describes what a player knows, or does not know, about her location in the tree when she has to act.
- Payoff (or utility) represents a player's ranking over the possible outcomes of the game.